Accuracy Is Not the Objective: A Cross-Disciplinary Review and Global Study of Surrogate Pipelines for Expensive Engineering Analyses

Working draft — review part

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Abstract

Draft. Surrogate models stand in for expensive simulations inside design loops. Engineering practice is dominated by Gaussian-process (kriging) surrogates, while tabular machine learning and several simulation-heavy sciences are dominated by tree ensembles; a thesis finite-element study found gradient-boosted trees out-predicting every Gaussian-process variant on real data while losing on smooth synthetic surfaces. This paper reviews surrogate, uncertainty, sampling, multi-fidelity, change-of-basis, upscaling and surrogate-based-optimization methods across disciplines, from computation theory to engineering practice, and then (in the study part, in preparation) scores complete pipelines by the true-function regret they deliver per unit of simulation cost.

1 Introduction

Design and analysis loops—optimization, uncertainty propagation, calibration, reliability—call a simulation many thousands of times. When one call costs minutes to hours, as for the coupled finite-element analyses behind the present work, the loop is run on a *surrogate*: a cheap model fitted to a limited set of simulation results [1–3]. The idea has been reinvented under many names in many fields—response surfaces in chemical engineering [4], kriging in mining geostatistics [5–7], design and analysis of computer experiments in statistics [8, 9], proxy models in petroleum engineering [10], emulators in climate science, cosmology and epidemiology [11–13], metamodels in simulation and operations research [14], surrogate benchmarks in automated machine learning [15], and fitness approximation in evolutionary computation [16, 17].

Each community has converged on its own default. In engineering design the Gaussian process (GP) is the dominant surrogate [2, 18], and multi-fidelity GP formulations are the natural extension when cheaper analyses are available [19–21]. In tabular machine learning, tree ensembles remain state of the art at small-to-medium sample sizes [22–24], and tree-based surrogates are preferred for algorithm-configuration benchmarks [15, 25] and agent-based model calibration [26]. These defaults are rarely confronted with each other on the same problems, with the same budgets, under the same noise.

The confrontation is not academic. The surrogate study of Poznański [27, Sec. 14.3 and App. G], which sized the deployable duct wall of a tube-launched ramjet from coupled fluid–structure finite-element analyses, reported a GP with the lowest error on six synthetic surfaces, yet gradient-boosted trees with the lowest cross-validated error on the real finite-element data; the trees were set aside because the downstream interior-point optimizer requires smooth derivatives. That study selected its

surrogate by approximation accuracy, implicitly assuming that the most accurate surrogate yields the best design—an assumption contradicted by several benchmarks in which the approximation winner and the optimization winner differ [15, 28, 29]. The present paper revisits and extends that study; where its findings are corrected here, the corrections are stated explicitly.

This paper has two parts. The **review** (Sections 2–9) surveys, breadth first and across disciplines, the methods that make up a surrogate pipeline: the learner (Section 2), its uncertainty estimate and the sampling it drives (Section 3), multi-fidelity fusion (Section 4), change-of-basis and latent-space representations borrowed from data-driven reduced-order modelling (Section 5), the conversion of a learner into an optimizer-ready representation (Section 6), and surrogate-based optimization itself (Section 8), followed by the theory that predicts when each should win (Section 9). We deliberately include old methods that were not pursued and methods that are standard outside aerospace but rare within it. The **study** (in preparation) varies surface class, dimension, budget, noise level and type, fidelity structure and cost, scout–learner pairing, upscaling and optimizer type in a factorial design, and scores pipelines by true-function regret per unit simulation cost. Section 11 states the gap the study fills.

2 The landscape of surrogate models

2.1 Lineages

Surrogate families descend from five traditions, and each inherited its home field's assumptions.

Geostatistics → kriging → Gaussian processes. Krige's regression corrections for gold-grade estimation on the Witwatersrand [5] were formalised by Matheron as optimal linear prediction of a random function [6]; Cressie traces the distinction between the two [7]. Sacks, Welch, Mitchell and Wynn transplanted the method to deterministic computer codes, treating the code output as a realisation of a stochastic process [8], which founded the design and analysis of computer experiments [9, 14, 18]. The same predictor is the GP regression of machine learning [30]. Extensions address trends (universal and blind kriging [31]), gradients [32], high dimension through partial least squares [33], input-dependent and stochastic noise [34, 35], scale [36, 37], and non-stationarity by warping, partitioning or depth [38–40]. The treed GP was itself motivated by a rocket-booster CFD experiment [39].

Approximation theory → splines, radial basis functions, sparse grids, rational functions. Hardy's multiquadric for topographic surfaces [41] and Duchon's thin-plate splines [42] began scattered-data approximation, now a mature theory with native-space error bounds [43–45]; compactly supported kernels make it sparse [46]. Tensor-product B-splines [47] give smooth, cheap, differentiable representations on structured grids, at a cost exponential in dimension; Smolyak's construction [48, 49] breaks that cost for functions of mixed smoothness. Chebyshev technology [50] and rational approximation, revived by the AAA algorithm [51], offer spectral accuracy for analytic functions and natural representation of pole-like features such as the $1/R$ collapse knee of our application.

Statistics → regression, splines and additive models. Response-surface methodology [4] fits low-order polynomials; sparse regression [52] selects terms from large dictionaries, as in ALAMO [53] and, for dynamics, SINDy [54]. Smoothing splines are the GP posterior mean for a particular prior [55, 56]; penalised B-splines (P-splines) fit splines directly to scattered, noisy data [57];

additive models and thin-plate regression splines scale them up [58, 59]; shape constraints encode monotonicity [60, 61]. Friedman's MARS [62] builds adaptive hinge bases that represent kinks exactly, and projection pursuit regression [63] sums ridge functions of learned projections—an early learned change of basis (Section 5).

Machine learning → trees, ensembles, neural networks, foundation models. Regression trees [64] partition the input space into boxes; bagging them gives random forests [65, 66], boosting them gives gradient-boosting machines [67] and their engineered descendants [68–70]; Bayesian sums of trees give BART [71], which has been used with sequential design for computer experiments [72]. Linear-leaf model trees [73] and Mondrian forests [74] are less common variants. Neural surrogates range from shallow perceptrons to deep ensembles [75], Bayesian networks [76–78], derivative-aware Sobolev training [79], Kolmogorov–Arnold networks with spline edges [80] and, most recently, tabular foundation models that perform Bayesian inference in context [81]. Physics-informed networks and neural operators [82–84] target field outputs and are outside our scalar-response scope. The polynomial neural networks of Ivakhnenko's group method of data handling [85]—self-organising, validation-selected polynomial networks from the late 1960s—are a largely abandoned branch of the same tree.

Physics and numerical analysis → spectral expansions and mapped models. Polynomial chaos [86, 87] expands the response in orthogonal polynomials of the inputs; sparse variants recover it from few runs [88]. Space mapping [89, 90] aligns a cheap physics model with an expensive one rather than fitting data from scratch (Section 4).

Simple local methods—Shepard's inverse-distance weighting [91], Sibson's natural-neighbour interpolation [92], moving least squares [93], LOESS [94] and Lipschitz interpolation [95]—are mostly absent from engineering comparisons but are the right baselines: as the climate-emulation literature shows, simple methods can match or beat large models when noise is significant [96].

2.2 What each field optimises for

The disagreement between fields is partly explained by what they benchmark on. Engineering comparisons [97–103] typically use smooth analytic functions, small samples, and deterministic outputs; GPs and RBFs win there, as they did in the synthetic-surface study of [27, App. G]. Williams and Cremaschi trained 16 632 models of eight families on 99 functions of the Surjanovic–Bingham library across five shape classes and found the winner depended on shape: GP most often on valley, bowl and miscellaneous shapes, MARS on many-local-minima functions; the best single predictor of accuracy was a spacing statistic of the design, not a landscape feature [28, 104]. In tabular machine learning, Grinsztajn et al. isolated three mechanisms behind the trees' advantage: neural networks are biased toward overly smooth functions (smoothing the training target hurt trees and barely affected networks), are harmed more by uninformative inputs, and are rotation-invariant where the data are not [22, 105, 106]. A stationary-kernel GP shares the first bias and, with an isotropic kernel, the third. In agent-based modelling, ragged simulator responses led Lamperti et al. to boosted trees precisely because kriging's smoothness assumptions "collapse interesting patterns" [26]. Process systems engineering, water resources, building energy, reliability, reservoir engineering, finance, turbomachinery and additive manufacturing each report their own mix [10, 107–116].

The common lesson of every large benchmark is that there is no universal winner but there are predictable ones [29, 88, 97, 104, 112, 117]. Two further lessons shape our design. Within-family configuration can matter as much as the choice of family—solver and sampler choices changed sparse-PCE errors by orders of magnitude [88]—so each family must be given its best configuration

under an equal tuning budget. And ensembles of surrogates weighted by cross-validation error [118–120] hedge against picking the wrong family when the predictors of the right one are unavailable.

3 Uncertainty estimates and the sampling they drive

3.1 Uncertainty estimators by family

Adaptive sampling needs a surrogate that knows where it is wrong. GPs provide a posterior variance in closed form [30]; note, however, that for fixed hyperparameters this variance depends only on the input locations, not on the observed outputs, so a greedy maximum-variance rule is a space-filling design in disguise until hyperparameters are re-estimated. Other families need constructed estimates: bootstrap committees and query by committee [121]; the jackknife and infinitesimal jackknife for random forests [122]; quantile regression forests [123]; distributional boosting [124] and virtual ensembles for boosting [125]; Bayesian sums of trees [71]; deep ensembles [75], Monte Carlo dropout [126], Laplace approximations [78] and evidential heads [127, 128] for networks [129]; and model-agnostic conformal intervals [130]. Tabular foundation models return predictive distributions directly [81]. These estimators differ in calibration and, importantly for sampling, in *shape*: a tree committee's spread is piecewise constant, so its maximiser is degenerate over whole cells.

3.2 Initial designs

Space-filling designs—Latin hypercubes [131], maximin designs [132], low-discrepancy sequences [133, 134] and centroidal Voronoi tessellations [135]—bound integration error through discrepancy and approximation error through fill distance [43]. Several large studies found the choice among them second-order: Williams and Cremaschi saw no significant difference between Halton, Sobol' and LHS [104], and in high dimension no advanced scheme beat LHS for sparse PCE [88]. The more consequential comparison is space-filling versus *independent random* sampling, which is the control most adaptive-sampling studies report, including [27, App. G].

3.3 Adaptive sampling for global accuracy

Reviews of adaptive sampling for global metamodelling [136–138] and a replicable comparison for kriging [117] catalogue the main criteria: maximum variance and active learning [139]; integrated variance reduction [140]; mutual information [141]; expected improvement for global fit [142]; expected prediction error [143]; local-nonlinearity-weighted Voronoi exploration [144]; Taylor-expansion error indicators [145]; and cross-validation-based criteria [146]. Fuhg et al. conclude that success depends on the problem and the goal of the analysis [117]. Moustapha et al. make the same point for reliability, where 39 combinations of surrogate, reliability estimator, learning function and stopping rule were run on 20 problems: no strategy dominated, but feature-dependent recommendations emerged [112]. Their modular decomposition is the template for our stage-swap ablation.

3.4 Sampling for a goal

When the goal is an optimum rather than a global fit, acquisition functions trade exploration for exploitation: probability of improvement [147], expected improvement and EGO [148], confidence bounds with regret guarantees [149], knowledge gradient [150], entropy search variants [151, 152],

constrained variants [153], trust-region schemes for high dimension [154], random-forest and density-ratio surrogates [155, 156], and stochastic RBF methods [157, 158]. When the goal is a boundary, as for reliability or for the feasible/ infeasible and floor-bound regions of the duct-sizing problem of [27, Sec. 14.3], contour-targeting criteria apply [159, 160]. A pipeline that samples for global accuracy and then optimizes—the pipeline of [27, Sec. 14.3]— therefore spends part of its budget where the optimizer will never go; whether that matters is an empirical question we test.

4 Multi-fidelity modelling

Most engineering analyses come with cheaper approximations: coarser meshes, lower-order physics, closed-form sizing rules. Peherstorfer, Willcox and Gunzburger classify methods that exploit them as *adaptation*, *fusion* and *filtering*, and stress keeping the high-fidelity model in the loop for guarantees [161, 162]; other reviews organise the surrogate side [163, 164].

Correction (bridge) functions. The oldest approach scales or shifts the low-fidelity model: $f_H \approx \beta(x) f_L + \delta(x)$, with β from sensitivities [165], kriging [166] or trust-region model management that enforces first-order consistency [167, 168]. The regression target of [27, Sec. 14.3, App. G]—the ratio of finite-element to closed-form thickness—is exactly a multiplicative bridge, an unnamed multi-fidelity model.

Autoregressive co-kriging. Kennedy and O'Hagan model each level as a scaled previous level plus an independent GP discrepancy [19]; Forrester, Sóbester and Keane brought it to optimization [20]; Le Gratiet and Garnier showed a recursive formulation gives the same predictor with independent, cheaper fits [146]; Han and Görtz's hierarchical kriging uses the low-fidelity predictor as the trend [169].

Nonlinear fusion. When fidelities are related nonlinearly, NARGP feeds the low-fidelity prediction into the high-fidelity kernel [170], deep GPs generalise it [171], and composite neural networks learn linear and nonlinear correlations in parallel [172]. Space mapping aligns inputs instead of outputs [89, 90]. For tree ensembles and generic learners, the simplest fusion—adding the low-fidelity prediction as an input feature—is natural but, to our knowledge, rarely benchmarked against co-kriging.

When it helps. Toal's guidance for multi-fidelity kriging is that fidelities must be strongly correlated (squared correlation around 0.9) and that roughly 10–80% of the budget should go to low fidelity [173]. Van Rijn and Schmitt show that correlation alone is a poor guide and propose "error grids" over the numbers of high- and low-fidelity samples, estimated by subsampling a single design [174], using the benchmark collection of [175]. The NATO AVT-354 group proposes analytical benchmarks spanning localised, multimodal, discontinuous and noisy behaviour together with global-accuracy and goal-oriented metrics [176, 177]; cost-aware multi-fidelity acquisition extends Bayesian optimization [178]. In hypersonic vehicle design, multi-fidelity GP surrogates with trajectory-informed and adjoint-guided sampling make coupled aero-thermal-trajectory optimization tractable [21, 179].

What is missing is a comparison of multi-fidelity methods *across learner families*, under matched cost, as a function of fidelity correlation, relationship type, cost ratio and budget split—and on a real problem where the low-fidelity model is available everywhere, as the closed-form sizing rule is in our finite-element data.

5 Changing the domain: latent, sparse-basis and low-rank surrogates

Data-driven reduced-order modelling succeeds by finding a basis in which dynamics are simple: proper orthogonal decomposition and dynamic mode decomposition find low-rank linear structure, SINDy finds a sparse dictionary representation [54], and the SINDy autoencoder learns the coordinates and the sparse model jointly [180]. Compressed and randomised DMD do not remove any stage; they compute the same decomposition from a sketch of the data, with cost set by the intrinsic rank rather than the ambient size [181–183]. The same ideas apply to static response surfaces, where they are scattered across communities that rarely cite each other.

Linear input reduction. Projection pursuit regression sums smooth functions of learned projections [63]; active subspaces take the dominant eigenvectors of the expected gradient outer product [184]; polynomial ridge approximation fits the subspace and the polynomial jointly without gradients [185]; GPs with built-in dimension reduction learn the projection inside the kernel [186]; random embeddings exploit low intrinsic dimension for optimization [187].

Co-learned nonlinear maps. Deep kernel learning places a GP on the output of a neural feature map and trains both by marginal likelihood [188]—the most direct form of "learning the domain and the fit together". Its weakness is decisive at engineering sample sizes: with many hyperparameters the marginal likelihood prefers to correlate all points, and the method can over-fit worse than a plain network unless treated fully Bayesianly [189]. Kernel flows learn kernels by a cross-validation-like criterion instead [190]; input warping restricts the map to per-axis monotone transforms [38]; Fourier features and spectral-mixture kernels learn spectral content [191, 192]. Latent-space optimization with variational autoencoders works for structured inputs [193], but an autoencoder trained only on uniformly sampled inputs learns the sampling density, not the response; for static surrogates the map must be supervised.

Sparse bases and compressed sensing. If the response is sparse in a known basis, ℓ_1 recovery needs samples in proportion to the sparsity rather than the basis size [194, 195]; for orthogonal polynomial bases this yields compressive-sampling polynomial chaos [196–198], and wavelets do the same for localised features [199]. The guarantees need random samples from the right measure, which couples the basis to the sampling design.

Low rank: the singular value decomposition of a surface. A bivariate function admits a Schmidt (functional SVD) expansion, the basis of two-dimensional Chebyshev technology [200]; in higher dimension the tensor-train format stores a function with cost linear in dimension and learns it by cross approximation from adaptively chosen samples [201, 202], with continuous analogues for surrogates [203]. Randomised Tucker and tensor-train decompositions compress structured grids [204]; Nyström and random-feature approximations compress kernel matrices [205, 206]; sketching compresses least squares [207]. These are the static counterparts of compressed DMD: they accelerate an intermediate stage—for us, densification of the intermediate model onto a grid, and repeated refits of the scout—without changing what is computed.

Automatic co-search. Choosing a transform, a basis and a learner together is an instance of the combined algorithm selection and hyperparameter optimization problem of automated machine learning [208, 209]; its selection bias must be controlled by nested validation.

6 From learner to optimizer-ready surrogate

A gradient-based optimizer such as an interior-point method [210] needs a surrogate that is cheap to evaluate and has well-behaved first and second derivatives. Many accurate learners fail one requirement: a GP's prediction costs $O(n)$ per point and its derivatives must be assembled; trees and forests are piecewise constant with zero gradient almost everywhere. A common engineering remedy, and the production pipeline of [27, Sec. 14.3], is a two-stage construction: fit an intermediate model to scattered, adaptively placed samples, densify it onto a structured grid, and interpolate the grid with tensor-product B-splines [47]. Poznański [27, App. G] found the grid–spline step preserved GP accuracy and cut evaluation cost 19–32-fold, but tested a single grid resolution and interpolant.

The literature offers several alternatives with different error and cost profiles. Smoothing splines and P-splines can be fitted directly to scattered noisy data, removing the intermediate model [56, 57, 59]; multilevel B-spline approximation builds a C^2 surface from scattered data by coarse-to-fine lattice refinement [211]; hierarchical and truncated hierarchical B-splines refine locally without a full tensor grid [212, 213]; Smolyak sparse grids reduce the grid size for mixed-smooth functions [48, 49]; Chebyshev interpolation is spectrally accurate for analytic intermediates [50]; low-rank compression of the densified grid (Section 5) reduces storage and evaluation cost from exponential to linear in dimension [201, 204]. The error of any two-stage construction decomposes into the intermediate model's error plus the upscaling error; the second vanishes at the spline's approximation order only when the intermediate model is smooth. For a piecewise-constant intermediate such as a tree ensemble, refining the grid reproduces the staircase more faithfully—which may help accuracy and hurt a gradient-based optimizer at the same time, whereas a *smoothing* upscaler acts as a regulariser. Whether upscaling can turn the most accurate but non-differentiable learner into the best optimizer-facing surrogate is, to our knowledge, untested.

7 The surrogate–optimizer interface

Surrogates are compared almost everywhere by how well they approximate the function. The optimizer, however, never sees "the surrogate" in the abstract: it sees one particular presentation of a trained model—its mean, a sample, a smoothed version, a risk-adjusted value, an exact mixed-integer encoding—and it receives, or does not receive, the model's gradients. It also searches exactly where the model is least trustworthy, because it is drawn to the points the model scores best. These choices form an *interface* between the trained model and the optimizer. Each facet has its own literature, mostly in separate communities; none of them is a standard axis of surrogate benchmarks, and Section 8 shows that accuracy alone is a weak guide to optimization outcomes. This section reviews the four facets in turn: what the optimizer sees (ensembles and stochastic intermediate models), whether the model's structure is used for exact global optimization, how the optimizer exploits the model's errors, and which gradients it uses.

7.1 What the optimizer sees: ensembles and stochastic intermediate models

Section 6 treats the intermediate model as one deterministic function to be smoothed. A second line of work keeps *many* models, or a random one, and lets the optimizer see the spread between them. Three uses can be separated: averaging to reduce error, randomness to drive exploration, and stochastic gradients that let first-order methods leave poor basins.

Averaging ensembles. In engineering design the ensemble of surrogates is a weighted average of different model classes, with weights from cross-validation error: the PRESS-weighted average

surrogate [118], weights optimised to minimise the ensemble's cross-validation error [214], and the finding that most of the benefit comes from generating many diverse surrogates and selecting by PRESS, with weighting gains that shrink in high dimension [119]. A recent review [215] reports ensembles as the usual remedy for not knowing which surrogate is best, which is the selection problem of this paper seen from the averaging side. In machine learning the ensemble is homogeneous and randomised: bagging fits one class to bootstrap samples [216], and subsampling without replacement (subagging) has the same variance-reduction mechanism with a cleaner analysis [217]; random forests add feature subsampling [65]; deep ensembles train networks from different initialisations [75]; Monte-Carlo dropout samples sub-networks [126]; and randomised prior functions add a fixed random network to each member so that uncertainty exists away from the data, which plain bootstrapping lacks [218]. All of these give a mean predictor and a spread; the spread is what acquisition rules (Section 3) consume.

Randomness for exploration. The member-to-member variation can replace a posterior. Thompson sampling optimises one random draw of the model per step; with ensembles the draw is one member, and the same idea underlies trust-region Thompson sampling in TuRBO [154]. Multiple-surrogate EGO adds one point per surrogate per cycle and imports uncertainty estimates from one surrogate to another, so that models without native uncertainty can be used [219]; bootstrapped kriging replaces the plug-in kriging variance in expected improvement [220]; ensembles of Gaussian processes with different kernels are combined in a mixture posterior for Thompson sampling [221]; and ensemble Bayesian optimization partitions the input space with random Mondrian trees and fits a GP per cell [222]. Neural ensembles are a common alternative to GPs in Bayesian optimization [223], but a systematic comparison found deep ensembles explore poorly and level off below other surrogates on standard benchmarks [224]. Tree ensembles are the surrogate of SMAC [155]. In evolutionary computation, surrogate-assisted memetic algorithms use ensembles and smoothed surrogates inside local search [17, 225], and the stochastic RBF method chooses each new point from random candidates scored by RBF prediction and distance, with probabilistic convergence to the global minimum [157]. For noisy simulators, stochastic kriging separates simulation noise from response-surface uncertainty [35], the regime where an averaged intermediate model is most defensible.

Stochastic gradients and smoothing. A deterministic gradient method stops in the first basin. Two mechanisms change that without a new surrogate. First, noise in the gradient: Langevin dynamics adds Gaussian noise to stochastic gradient steps and has non-asymptotic guarantees of reaching approximate global minimisers of non-convex objectives [226]; using one random ensemble member's gradient per step is a model-uncertainty version of the same noise. Second, smoothing: convolving a non-smooth function with a Gaussian or uniform kernel gives a smooth surrogate whose gradient can be estimated from random perturbations [227, 228]. For a piecewise-constant tree ensemble this is the only way to obtain useful gradient information, and a smoothing upscaler (Section 6) is its deterministic counterpart. Particle methods make the optimizer itself an ensemble: consensus-based optimization moves particles towards a weighted consensus point with an exploration noise [229] and converges globally under mild conditions [230], and ensemble Kalman inversion is a derivative-free ensemble optimizer from inverse problems [231].

Sums of local models. A surrogate that is a sum of simple local pieces is both smooth and naturally stochastic. The partition-of-unity method blends local approximations with weights that sum to one [232]; SNOBFIT fits local quadratic models around the current best points and branches

the domain to keep a global search [233]. Drawing one local piece by its weight gives a stochastic gradient whose expectation is the weighted local gradient. We found no study that uses such a decomposition to run stochastic first-order methods on a fitted surrogate.

7.2 Using the model's structure: exact global optimization of trained models

A trained model is not a black box to the optimizer that owns it. A tree ensemble is piecewise constant on a finite partition, so its global minimum over a box is a mixed-integer linear program [234]; ENTMOOT solves this with a distance-based uncertainty term for Bayesian optimization over gradient-boosted trees [235], and tree-ensemble kernels give trees a GP-like posterior whose acquisition is a mixed-integer second-order cone program [236]. A ReLU network is piecewise linear and can be encoded the same way [237], with ideal formulations that tighten the relaxation considerably [238]. Smooth models admit deterministic global optimization by branch-and-bound with McCormick relaxations propagated through the trained network [239] or through the Gaussian-process predictor and its acquisition functions [240]; OMLT automates such encodings of trained networks and gradient-boosted trees inside larger optimization models [241]. These methods certify the surrogate's global optimum, which local and stochastic searches cannot. Whether a *certified* optimum of the surrogate is a better design than a heuristic one depends on the surrogate's error near that optimum—the subject of the next facet—and has not, to our knowledge, been measured against the true function across model classes.

7.3 Exploiting the model's errors: the optimizer's curse and validity domains

Choosing the best of many estimated values selects positive errors: even with unbiased estimates the chosen option's realised value falls short of its estimate, the optimizer's curse [242]. A surrogate optimizer is the extreme case, since it searches the whole design space for the most favourable prediction and is drawn to regions where the model errs in its favour, typically away from the data. Offline model-based optimization, where no new evaluations are possible, treats this as the central difficulty: conservative objective models train the surrogate to under-estimate out of distribution before optimizing it [243], conditioning by adaptive sampling restricts the search to inputs a generative model deems realistic [244], and Design-Bench collects tasks on which naive gradient ascent on a learned model fails [245]. In process engineering the same concern leads to explicit validity domains: a one-class classifier fitted to the training inputs, after topological data analysis finds holes in the data, is added as a constraint to the optimization [246]; ENTMOOT's distance term plays a similar role for trees [235]. Risk-averse objectives (optimising $\mu + \kappa\sigma$ instead of μ) are the uncertainty-aware counterpart for surrogates that have a predictive variance. The engineering surrogate literature that this paper extends rarely reports the gap between the surrogate's value at its optimum and the true value there, although it is the quantity that decides whether the surrogate's optimum can be trusted.

7.4 The gradients the model offers

Gradient-based optimizers need derivatives. A surrogate can supply its own analytic gradient (kriging, polynomial chaos, networks through automatic differentiation) or be differentiated by finite differences inside the evaluation budget; for trees the analytic gradient is zero almost everywhere. Which source to use is usually decided by convenience. It interacts with the other facets: an analytic gradient of an accurate model may still point at a spurious minimum, and a smoothed or sampled gradient trades bias for exploration.

7.5 State of the art and the class of gaps

Each facet is mature within its own community: ensembles for accuracy and exploration (engineering design, Bayesian optimization, evolutionary computation), exact optimization of trained models (mathematical programming and process systems), model exploitation and validity (offline model-based optimization and process engineering). What is missing is the interface itself as an object of study: for one fitted surrogate, a matched comparison of the ways an optimizer can consume it, scored by regret on the true function and by the gap between promised and delivered values, across model classes and landscapes. Concretely: (i) averaging versus sampling members versus smoothing versus exact mixed-integer optimization of the same ensemble; (ii) whether member-sampling noise helps stochastic first-order methods relative to an equal amount of isotropic noise; (iii) whether guarding against extrapolation, or optimising a pessimistic objective, reduces true regret or only the surrogate's over-promise; (iv) whether the answers depend on the landscape type, as the accuracy results of this paper suggest they will.

This paper's study addresses the class rather than one item. The fit-once optimization study scores every optimizer on every surrogate with both gradient sources and records the surrogate's value at each pick; the upscaling study compares deterministic, sparse-grid, low-rank and stochastic presentations (a partition of unity of local quadratics, and a subsample ensemble of smooth members, each with a stochastic-gradient arm) of the same intermediate model; the surrogate-managed study adds bootstrapped-kriging EGO to the loops that return to the true function; and a dedicated interface study applies to one fitted surrogate the mean with local and global optimizers, Thompson sampling over ensemble members, member-sampled stochastic gradients against a Langevin-noise control, Gaussian-smoothing gradients, consensus-based optimization, a distance-guarded objective and a pessimistic objective, with optimised-weight and randomised-prior ensembles among the intermediate models [214, 218]. Exact optimization covers all three structured model classes with open-source tools only: tree ensembles and ReLU-network ensembles as mixed-integer linear programs [234, 237], solved by HiGHS, and the Gaussian-process mean by a spatial branch-and-bound written for this study. Its lower bound on each box is the larger of an interval bound on every kernel term, the scheme of Franey et al. [247] for expected improvement, and a centred bound: the mean and gradient at the box centre minus a Taylor remainder bounded by Cauchy–Schwarz in the kernel's reproducing-kernel Hilbert space. The second bound shrinks with the square of the box size for Matérn-5/2 and squared-exponential kernels, and it keeps the cancellation between the large, mixed-sign weights of a nearly interpolating GP, which makes the interval bound alone too loose to certify even in two dimensions. Every exact solve is time-limited and records its certified gap, so the study separates certified from uncertified optima. The tightened (ideal) network formulations of Anderson et al. [238] and the McCormick relaxations of Schweidtmann et al. [240], which bound the posterior variance and acquisition functions as well as the mean, are not reimplemented.

8 Surrogate-based optimization and the accuracy fallacy

Optimizers. Once a surrogate exists, any optimizer can be run on it: gradient-based interior-point and SQP methods [210, 248, 249]; derivative-free local methods [250–252]; deterministic global search [253]; and stochastic global methods [254–257]. Surrogate-managed methods instead return to the true function: EGO [148], trust-region model management with provable convergence [167], SNOBFIT [233], stochastic-RBF methods [157, 158], SMAC and TPE [155, 156], and TuRBO [154]. Surrogate-assisted evolutionary algorithms form a parallel literature [16, 17].

Accuracy is not the objective. Several independent results show that the most accurate surrogate is not reliably the one that produces the best decision. Eggensperger et al. found gradient boosting gave the lowest RMSE while random forests gave the highest rank correlation on all nine datasets, and that tree surrogates reproduced optimizer behaviour where GP surrogates sometimes did not [15, 25]. Williams and Cremaschi found MARS and GP best for approximation but random forests, SVR and GP most robust for surrogate-based optimization [28]. Loshchilov et al. use a surrogate that only ranks candidates, because comparison-based evolution strategies need nothing else [258]. Neufang et al. observed optimizer rankings change between test functions and chemical-process cases [29], and Lütjens et al. found the ranking of climate emulators flipped with the noise level of the training data [96]. Theory explains why (Section 9): the regret of the surrogate's optimum is bounded by the surrogate error on the near-optimal sublevel sets, not by its global mean-squared error, and gradient-based optimizers additionally depend on derivative accuracy.

Benchmarks. Test-function libraries—the Surjanovic–Bingham virtual library used here, the 175-function survey of Jamil and Yang, and the COCO/BBOB platform [259, 260]—classify functions by shape, modality, separability and conditioning; exploratory landscape analysis computes such features from samples [261, 262] and supports automated algorithm selection. Benchmarking practice—performance and data profiles, multiple-problem statistics and their Bayesian alternatives [263–267]—and surrogate-based benchmark suites [268] complete the toolkit.

9 What theory predicts

No free lunch, but structured lunches. Averaged over all functions, no optimizer or learner beats another [269]; every advantage comes from matching an inductive bias to a function class. Information-based complexity makes the price of generality explicit: for functions of smoothness s in d dimensions, the minimax error from n samples scales as $n^{-s/d}$ for approximation and $n^{-s/(2s+d)}$ for noisy regression [270, 271]. Smoothness and dimension, not the method's name, set the achievable rate.

Kernels and GPs. For kernel interpolation the error is controlled by the fill distance h of the design through the power function, at a rate set by the kernel's smoothness, with a trade-off between accuracy and conditioning [43, 44]; GP posterior contraction rates match when the prior's smoothness matches the truth's and degrade otherwise [272]. The fill-distance dependence explains why a spacing statistic of the design predicted accuracy in a large empirical comparison [104], and why space-filling designs are hard to beat for global fit. Expected improvement converges at known rates for functions in the kernel's reproducing-kernel Hilbert space [273], and confidence-bound acquisition has regret bounds [149]—guarantees that hold only when the function is as smooth as the kernel assumes.

Trees and networks. A piecewise-constant partition of n cells approximates a Lipschitz function at rate $n^{-1/d}$, slower than a well-matched kernel on smooth functions but indifferent to discontinuities, plateaus and floors; ensembles are consistent under mild conditions [66]. Networks escape the curse of dimension for functions with suitably bounded spectral norms [274] and approximate smooth functions efficiently with depth [275], but trained networks are biased toward low frequencies [105], and rotation-invariant learners pay linearly in the number of irrelevant inputs [106].

Noise. With observation noise of variance σ^2, any held-out error measured against noisy targets is bounded below by σ^2; differences between models shrink toward zero as σ grows, which is how a model-ranking study can be dominated by its noise level [96]. Clean-truth scoring on synthetic problems removes this floor.

From surrogate error to design quality. If $\hat{x}$ minimises the surrogate s and $x^\star$ minimises f, then $f(\hat{x}) - f(x^\star) \le 2 \sup |f - s|$ over the union of the near-optimal sublevel sets of f and s. Global mean-squared error neither bounds nor is bounded by this quantity; rank fidelity on the sublevel sets governs comparison-based optimizers, and derivative fidelity near $x^\star$ governs gradient-based ones. Multi-fidelity and compression methods enter the same bound through their effect on $\sup |f - s|$ per unit cost.

10 Computational complexity

Asymptotic cost decides which methods are *admissible* at a given sample size, dimension and simulation cost, before accuracy is considered. Table 1 collects fit, storage and prediction costs for the families reviewed above, in the usual algorithm for each; Table 2 does the same for the representations used to make a learner optimizer-ready. Constants matter at engineering sample sizes, so the study also reports measured times.

Three walls recur. Exact GPs and dense RBF systems need $O(n^3)$ time and $O(n^2)$ memory [30, 43]—about 0.8 GB of kernel matrix at $n = 10^4$—which inducing-point, Nyström and random-feature approximations reduce to $O(nq^2)$ [36, 205, 206], and partitioned or local GPs reduce by working on subsets [37, 39]. Full tensor grids, and hence tensor-product spline upscaling, cost $O(m^d)$: 24 points per axis is 1.4×10^4 nodes in three dimensions, 1.9×10^8 in six and 6×10^{13} in ten; sparse grids [49] and tensor-train formats [201, 202] restore near-linear scaling in d under mixed smoothness or low rank. Delaunay-based interpolants (piecewise-linear, natural neighbour) grow as $O(n^{\lceil d/2 \rceil})$ in the worst case and are impractical beyond about six inputs. By contrast, tree ensembles fit in $O(n \log n)$ per tree and predict in time independent of n [65, 68, 69], and in-context tabular models replace training by $O(n^2)$ attention [81].

Loops multiply these costs. An active-learning loop that refits an exact GP after each of N acquisitions costs $O(N^4)$ unless the Cholesky factor is updated, and joint co-kriging over L fidelities costs $O((\sum_\ell n_\ell)^3)$ while the recursive formulation costs $O(\sum_\ell n_\ell^3)$ [146]. Randomised linear algebra [182, 183] reduces the cost of a decomposition to its intrinsic rank, which is how compressed DMD accelerates its intermediate step. For finite-element simulators costing an hour per run, every surrogate term is negligible up to $n \sim 10^4$; for cheap simulators these terms decide the winner.

11 Gap and contributions

The reviewed literature establishes that no surrogate family dominates, that winners are predictable from problem features, that within-family configuration matters, and that approximation accuracy is a weak proxy for decision quality. It does not, to our knowledge, contain a study that jointly varies (i) surface class and landscape features, (ii) dimension and sample budget, (iii) noise level and type, (iv) fidelity structure and cost ratio, (v) the pairing of the sampling scout with the deliverable model, (vi) the upscaling from learner to optimizer-ready surrogate, including compression, (vii) the optimizer, and (viii) the interface between them—how the optimizer consumes the trained model (its mean, a sampled member, a smoothed or risk-adjusted value, or an exact mixed-integer encoding), which gradients it uses, and how far the surrogate's promised optimum falls short of the true

Table 1: Asymptotic costs of surrogate learners. n training points, d inputs, P basis terms, s active terms, q inducing points or features, B trees, T boosting rounds, D depth, W weights, E epochs, K ensemble members, HR hyperparameter iterations $\times$ restarts.

Method	Fit time	Memory	Predict (mean / variance)
Polynomial RSM, $P = \binom{d+p}{p}$	$O(nP^2 + P^3)$	$O(nP)$	$O(P)$ / $O(P^2)$
Sparse regression, sparse PCE (OMP)	$O(snP)$	$O(nP)$	$O(s)$
Exact GP / kriging	$O(HR\,n^3)$	$O(n^2)$	$O(nd)$ / $O(n^2)$
Sparse GP, Nyström, random features	$O(nq^2)$	$O(nq)$	$O(q)$ / $O(q^2)$
Treed GP (leaves n_j)	$O(S\sum_j n_j^3)$	$O(\sum_j n_j^2)$	$O(n_j)$
RBF interpolation (dense)	$O(n^3)$	$O(n^2)$	$O(nd)$
IDW / k-NN	$O(dn \log n)$ (tree build)	$O(nd)$	$O(\log n)$–$O(nd)$
Delaunay linear / natural neighbour	$O(n^{\lceil d/2 \rceil})$ worst case	same	$O(\log n)$
MARS, M terms	$O(ndM^2)$	$O(nM)$	$O(M)$
Random forest	$O(Bd'n \log n)$	$O(Bn)$	$O(BD)$
Gradient boosting (histogram)	$O(Tnd)$	$O(nd + T2^D)$	$O(TD)$
BART, m_t trees, S sweeps	$O(Sm_tn)$	$O(m_tn)$	$O(Sm_tD)$
MLP / deep ensemble	$O(KEnW)$	$O(KW)$	$O(KW)$
Deep kernel learning	$O(E(n^3 + nW))$	$O(n^2 + W)$	$O(nd + W)$ / $O(n^2)$
TabPFN (in-context)	—	$O(n^2)$	batched $O(nM)$
Tensor train (cross)	$O(dmr^3)$; $O(dmr^2)$ samples	$O(dmr^2)$	$O(dr^2)$
Compressed sensing (OMP), basis N	$O(snN)$	$O(nN)$	$O(s)$

Table 2: Costs of optimizer-ready representations; m nodes per axis, spline order k ($k = 4$ cubic), level λ, rank r.

Representation	Build	Storage	Evaluate
Tensor B-spline on dense grid	m^d model calls $+ O(dm^d)$	$O(m^d)$	$O(k^d)$
Multilevel B-spline (MBA)	$O(h(n + m_h^d))$	$O(m_h^d)$	$O(k^d)$
Smolyak sparse grid	$O(2^\lambda \lambda^{d-1})$ model calls	same	$O(2^\lambda \lambda^{d-1})$
Additive P-spline / GAM	$O(ndK)$	$O(dK)$	$O(dk)$
Tensor-train compressed grid	$O(dmr^3)$	$O(dmr^2)$	$O(dr^2)$

value (Section 7)—and scores complete pipelines by true-function regret per unit simulation cost, while explaining the persistent disagreement between engineering practice (Gaussian processes) and tabular machine learning (tree ensembles). The study part of this paper does this on the Surjanovic–Bingham functions, the multi-fidelity analytical benchmarks, and two real engineering datasets: coupled finite-element sizing of an inflatable ramjet duct, which carries a closed-form low-fidelity model at every point, and historical aircraft performance data.

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